Does AI Adoption Cause Occupational Downgrading?

A Research Design for Credential–Occupation Mismatch in High-Exposure Knowledge Work*

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* This draft revises my previous report in response to Prof. Hu's feedback. It narrows the project to a single core mechanism (occupational downgrading and credential mismatch) and proposes a unified empirical framework.

Contents

1 Research Motivation

The conventional diagnostic of AI displacement tracks aggregate unemployment among highly exposed workers, and read through that lens the early evidence is tentatively reassuring — unemployment among highly AI-exposed workers has not risen significantly (Massenkoff and McCrory, 2026). That observation is methodologically sound, but it speaks only to the *quantity* of employment. The diagnostic itself was honed on industrial-era technology shocks, and it may systematically miss the primary adjustment margin in knowledge work. I argue that it is a weak basis for inferring labor-market security, because the indicator may be misaligned with how knowledge workers actually absorb a task-substituting shock through a deterioration of job *quality* rather than quantity.

The task framework of Autor et al. (2003), extended by Acemoglu and Restrepo (2018, 2020), treats displacement as a quantity problem driven by position-eliminating substitution. Generative AI is plausibly a different kind of technology. Eloundou et al. (2023) document that exposure is concentrated among graduate-educated professionals, who possess the financial buffers and occupational flexibility that displaced manufacturing workers lacked. When such workers face task substitution, they may exercise that flexibility by moving into jobs that fall below their credential and experience level rather than into unemployment. And as generative models absorb routine analytical or drafting tasks, displaced professionals may step down into lower-tier support roles such as a legal associate shifting into paralegal work—to remain employed. This form of structural underemployment does not register in aggregate unemployment statistics, which were designed to capture outright job loss rather than the structural devaluation of skill and earnings within continued employment.

I term this possibility the *measurement–welfare mismatch*, defined as a condition in which the standard indicator reads stable while underlying welfare deteriorates. On this reading, stable unemployment figures in the early AI data need not indicate that adjustment has been benign. They may instead indicate that adjustment is operating through a channel the existing monitoring infrastructure was not built to observe. This project does not assume that interpretation is correct; it proposes a research design capable of testing it. I focus this test entirely on *occupational downgrading into credential–occupation mismatch*, as it is the single channel that is both first-order for welfare and cleanly measurable in existing panel data.

2 Research Question

Primary question: *Among initially well-matched, college-educated workers, does higher AI exposure of a worker's occupation raise the probability of transitioning into credential–occupation mismatch (employment in a job whose required education falls below the worker's* *attained credential) following the release of ChatGPT?*

The estimand is a within-worker treatment effect: the change, attributable to AI exposure, in the probability that a correctly matched worker downgrades into a below-credential job after the late-2022 shock relative to before it. A positive estimate would confirm that the absence of an unemployment signal documented by Massenkoff and McCrory (2026) is consistent with displacement operating inside employment rather than with displacement being absent. Alternatively, a precisely estimated null would act as a definitive economic result rather than a failed test. It would weaken the measurement–welfare mismatch hypothesis for this channel and redirect attention to the alternative adjustment channels discussed in Section 3.3.

3 Literature Review

3.1 Measurement Architecture and the Credential-Mismatch Gap

The standard diagnostic of AI displacement inherits a structural choice from the task-based framework of Autor et al. (2003), extended by Acemoglu and Restrepo (2018, 2020) to formalize competing displacement and productivity effects. Acemoglu and Restrepo (2022) establish the macroeconomic backdrop, showing that recent automation has been "so-so," delivering substitution without sufficient new-task creation to compensate—and that this dynamic accounts for much of the rise in US wage inequality over four decades. Both Eloundou et al.'s (2023) GPT substitutability scores and Massenkoff and McCrory's (2026) Observed Exposure measure nonetheless treat labor-market impact as a *quantity* problem: has employment among highly exposed workers declined? That framing fits position-eliminating substitution, as with industrial robots. It is less suited to a technology that operates through task substitution *within* knowledge roles, where graduate-educated workers hold the credentials and social capital to absorb displacement through a quality margin rather than a quantity margin. Autor (2019) provides a complementary mechanism, showing that technological change bifurcates "new work" into high-skill "frontier" jobs and low-skill "wealth work." This hollowing-out raises competitive pressure on mid-level workers and pushes some of the displaced into lower-tier roles rather than lateral ones.

The welfare cost of this quality margin is documented across four decades of overeducation research, and it motivates my choice of outcome. Leuven and Oosterbeek (2011) synthesize that literature and report a cross-study central tendency for the credential-mismatch wage penalty of roughly 15%—a loss that accrues without any unemployment event. Sicherman (1991) shows that overeducated workers face a systematic wage discount relative to matched peers and, absent promotion or job-to-job mobility, remain in suppressed-wage positions over extended careers. Furthermore, Pellizzari and Fichen (2013), using PIAAC data across OECD economies, find that skill mismatch is widespread and cross-nationally durable, with little evidence of systematic transition back into matched employment. This represents a structural sorting failure rather than frictional adjustment, creating a low-visibility trap that aggregate unemployment statistics cannot detect. Autor and Handel (2013) provide the direct bridge between the task and mismatch literatures, showing that occupational wages are governed by the task content of assigned roles rather than by credential level alone; credential–occupation mismatch can therefore suppress earnings through organizational task allocation, without any underlying deterioration of worker skill.

The gap I target is empirical rather than conceptual: to my knowledge, no study causally links the intensity of AI exposure in a worker's occupation to the credential–occupation match of the same worker's subsequent employment within a panel. Without this link, the Massenkoff and McCrory (2026) stability finding admits two readings that the data cannot yet separate—displacement is absent, or displacement is operating outside the boundaries of the measurement architecture. The design in Section 4 is built to separate them.

3.2 The Downgrading Cascade and Persistent Earnings Penalties

This is the mechanism the project tests, and the rest of the review is organized around it. Gibbons and Waldman (1999) formalize the career ladder as an assignment mechanism in which firms match workers to tiers based on revealed productivity signals, with promotion following when output crosses a tier threshold. Extending this logic to a market-wide shock yields a downward-cascade prediction. If AI compresses the employment capacity of a tier, the model predicts that workers above it need not absorb the loss in place; some compete downward. A displaced senior worker may take a mid-level position, the incumbent mid-level worker an entry-level position, and so on. Such crowding-out can leave employment counts stable across intermediate tiers while compressing wages and match quality. Kambourov and Manovskii (2009) show that human capital is substantially occupation-specific rather than firm- or industry-specific, which implies that each step down this cascade destroys a portion of a worker's accumulated skill—the welfare content of the mismatch outcome.

The earnings consequences of involuntary downward mobility are well established. Jacobson et al. (1993) show that workers displaced from high-tenure positions face earnings losses averaging around 25% with little recovery. Crucially, these losses are concentrated among workers who subsequently find employment, indicating that the loss reflects a permanent drop in job quality rather than unemployment duration. Davis and von Wachter

(2011) translate such deficits into lifetime present-value terms, with losses of roughly 1.4 to 2.8 years of pre-displacement earnings depending on aggregate conditions. However, the most directly relevant identification template is Autor et al. (2014). Exploiting cross-regional trade exposure, they document individual-level earnings losses that fall disproportionately on workers in the lower half of the wage distribution and operate through reduced within-year earnings rather than extended unemployment. Finally, Cortes (2016) uses panel data to trace individual transitions, finding that displaced middle-wage workers move systematically into lower-wage occupations—confirming that the cascade operates at the individual rather than the purely compositional level. The within-worker panel design I adopt follows this template directly.

Generative AI may amplify this cascade for a specific structural reason. Unlike the industrial automation studied by Acemoglu and Restrepo (2020), which targeted routine workers with limited downward amplitude, AI exposure is concentrated near the top of the occupational hierarchy. If a displacement cascade operates from this level, it has the greatest potential to disrupt the tiers below. Massenkoff and McCrory's (2026) finding that young workers in high-exposure fields face a 14 percentage-point decline in job-finding rates is consistent with a demand contraction at the entry tier; it does not, on its own, establish whether incumbents above that tier are downgrading. That is the question this design isolates.

3.3 Related Invisible-Adjustment Channels

Occupational downgrading does not exhaust the ways AI can erode welfare without moving the unemployment rate. At least two adjacent channels share the same measurement–welfare mismatch property, and I delimit the present design against them. Each turns on a different margin and demands its own data and identification strategy; pursuing all three at once would diffuse the contribution and blur the identification that motivates the design. I therefore outline the two channels below to map the broader landscape of hidden displacement, while keeping the empirical test strictly bounded to occupational downgrading.

Human-capital pipeline depletion: The economic rationale for firm-sponsored training is well established (Becker, 1964; Acemoglu and Pischke, 1998, 1999), with early-career human capital formation driving substantial seniority premia later in life (Topel, 1991; Neal, 1995). Entry-level roles are critical to this process because foundational skills such as structuring a client analysis or diagnosing a software failure, are acquired largely through learning-by-doing, making the junior rung the foundation for a worker's lifetime earnings. However, Brynjolfsson et al. (2023), studying 5,179 customer-support agents, find that generative AI raises novice productivity by codifying the tacit knowledge of top performers. This dynamic could erode the economic rationale for entry-level hiring, as Hershbein and Kahn (2018), Kahn (2010), and Deming and Noray (2020) show is costly when it occurs. What makes this mechanism distinct is its permanence. While macroeconomic downturns suppress junior hiring cyclically, the substitution of algorithmic capital for novice labor threatens a structural, prolonged decline in early-career demand. However, testing this channel requires data on the hiring of new labor-market entrants and a design centered on cohort entry rather than incumbent transitions; I therefore set it aside as an extension.

Contractual fissuring: Falling coordination and monitoring costs incentivize firms to convert salaried employment into alternative contract arrangements without triggering measured unemployment (Coase, 1937; Weil, 2014), a secular trend increasingly visible among college-educated workers (Katz and Krueger, 2019). Generative AI plausibly exacerbates this dynamic by making task boundaries legible. Roles previously bundled into a single position to solve a contracting problem can now be unbundled into modular pieces procured independently. And the costs of this transition are substantial. A former salaried analyst who moves to a contract role appears in no unemployment count yet forfeits employer-sponsored health insurance, retirement contributions, and the implicit income insurance of a durable relationship. Goldschmidt and Schmieder (2017) show the associated wage penalties are real (10–15% for the same tasks under domestic outsourcing), which is why the channel matters for welfare. However, because this fissuring channel operates through W-2 tax classifications and firm boundary data rather than credential–occupation mismatch, it constitutes a fundamentally separate empirical object requiring its own identification strategy.

The macroeconomic stakes that make all three channels worth monitoring are formalized by Hemenway Falk and Tsoukalas (2026), whose demand trap is activated by structural income deflation rather than by aggregate joblessness. While I do not intend to test their general equilibrium framework directly, their model provides the essential theoretical motivation for why closing this measurement gap matters for welfare.

4 Empirical Strategy

This section formalizes the empirical architecture of the project, restricting the design to a single outcome, treatment, and identification strategy. To isolate the effect of the technology, the framework exploits a common macroeconomic shock—the public release of ChatGPT in November 2022—interacted with continuous, predetermined variation in occupational AI exposure.

4.1 Outcome: Credential–Occupation Mismatch

For worker i employed in occupation $o(i,t)$ in month t, the primary outcome is a binary credential–occupation mismatch (overeducation) indicator,

$$\text{Mismatch}_{it} \;=\; \mathbf{1}\Big[\,\text{educ}_i \;>\; \text{ReqEduc}\big(o(i,t)\big)\,\Big]\,, \tag{1}$$

where educ_i is the worker's attained education and $\text{ReqEduc}(\cdot)$ is the education required by the occupation. I operationalize $\text{ReqEduc}(\cdot)$ using the realized-matches method: a worker is overeducated if attained schooling exceeds the occupation's modal schooling (Kiker et al., 1997) (robustness: mean plus one standard deviation, following Verdugo and Verdugo, 1989) among incumbents, with the benchmark fixed in a *pre-shock base year* (2021) so that the treatment cannot move the yardstick. Because the outcome is binary, the parameter of interest is a transition probability; restricting the estimation sample to workers who are correctly matched at baseline makes the estimand the hazard of *downgrading into* mismatch,

$$\Pr(\text{Mismatch}_{it} = 1 \mid \text{Mismatch}_{i,t-1} = 0,\ \text{Exposure}_i,\ \text{Post}_t)\,. \tag{2}$$

As a robustness outcome I replace the realized-matches benchmark with the required education implied by O*NET Job Zones[1] mapped to SOC codes. While coarser (five zones), this classification provides an exogenous, non-mechanical validation of the primary mismatch indicator.

4.2 Treatment: Occupational AI Exposure as a Continuous Dose

The treatment is the AI exposure of the worker's *baseline* occupation,

$$\text{Exposure}_i \;=\; \text{GPT-exposure score of worker } i\text{'s 2021 occupation } o \;\in\; [0,1], \tag{3}$$

measured with the GPT exposure scores of Eloundou et al. (2023)[2] and standardized to mean zero and unit variance, so that the coefficient of interest reads as the effect of a one-standard-deviation increase in exposure. Fixing exposure at the pre-shock (2021) occupation makes it predetermined: it cannot be an artifact of post-shock job switching, which is itself an outcome. I interpret the Eloundou score as *potential* exposure rather than realized adoption and treat the resulting estimate as an intention-to-treat parameter; as a robustness check I substitute realized usage shares from the Anthropic Economic

[1] Job Zones are distributed with the O*NET database. Available at: `https://www.onetcenter.org/database.html`.

[2] Exposure scores and replication materials available at: `https://github.com/openai/GPTs-are-GPTs`.

Index (Anthropic, 2025) as the dose.[3]

A feature of this dose bounds what β identifies: the Eloundou score captures *generative*-AI (large-language-model) exposure, not the predictive or automation AI that preceded it. The estimand is therefore the effect of the *generative*-AI shock on top of any earlier adjustment, not the total effect of AI. This distinction is not academic. Frank et al. (2026) document that risk in AI-exposed occupations began rising in early 2022—months before ChatGPT—so part of the deterioration I study predates the shock and is absorbed into the worker and pre-period baselines, which makes β a conservative, generative-AI-specific estimate rather than a measure of all AI-driven downgrading. Frank's finding cuts both ways; however, it also warns against treating the ChatGPT launch as a clean natural experiment, a challenge I confront directly in Section 4.5.

4.3 Identification: Common-Shock Dose-Response Difference-in-Differences

The static specification is

$$\text{Mismatch}_{it} = \alpha_i + \lambda_t + \beta\left(\text{Exposure}_i \times \text{Post}_t\right) + \mathbf{X}_{it}\boldsymbol{\gamma} + \varepsilon_{it}, \tag{4}$$

where:

- α_i are worker fixed effects. They absorb every fixed difference across workers, so β is identified purely from within-worker change over time.
- λ_t are calendar-month fixed effects. They absorb anything common to all workers in a given month, including the average impact of the ChatGPT release itself.
- $\text{Post}_t = \mathbf{1}[\,t \geq \text{Nov 2022}\,]$ marks the post-shock period.
- $\mathbf{X}_{it}$ collects time-varying controls such as age and, where available, tenure.

The calendar-month effects already absorb the common shock, so β is identified only from the *interaction* of exposure with the post-shock period. It captures how much more high-exposure workers downgrade after ChatGPT than low-exposure workers do over the same months. Under the identifying assumption stated below, that difference is causal, giving the effect of a one-standard-deviation increase in baseline AI exposure on the probability of downgrading into credential mismatch after the shock relative to before. Without that assumption it remains only a conditional association. I estimate the equation as a linear probability model with the two-way fixed effects, and I cluster standard errors at the occupation level, where the treatment varies. Because that leaves

[3] Available at: `https://www.anthropic.com/economic-index`; the underlying data are released at `https://huggingface.co/datasets/Anthropic/EconomicIndex`.

relatively few clusters, I confirm inference with a wild-cluster bootstrap and evaluate the parallel trends assumption via a formal joint significance test of the event-study leads, rather than relying on visual inspection.

The dynamic counterpart both tests the identifying assumption and produces the figure that carries the paper,

$$\text{Mismatch}_{it} \;=\; \alpha_i \;+\; \lambda_t \;+\; \sum_{k\neq -1} \beta_k \left(\text{Exposure}_i \times \mathbf{1}[\,t-t^* = k\,]\right) \;+\; \varepsilon_{it}, \qquad t^* = \text{Nov 2022.} \tag{5}$$

The leads β_k for $k < 0$ should be statistically indistinguishable from zero; this is the falsifiable form of the parallel-trends assumption. The lags β_k for $k \geq 0$ trace the dynamic downgrading response.

Adjustment in knowledge-intensive labor markets is rarely instantaneous. Search frictions, fixed-term and annual contracts, and organizational inertia all delay the labor-market bite of the shock. For this reason I do not expect an effect in the event month. The response is more likely to emerge with a lag of roughly two to four quarters, once firms re-architect workflows and revise credential requirements. Coefficients at small k may therefore be near zero even if the hypothesis holds, which sharpens rather than weakens the test. The decisive window is about a year out, and a path that stays flat over it would itself be evidence against the mechanism.

Identifying assumption: Absent the ChatGPT shock, the mismatch-transition probability of high- and low-exposure workers would have evolved in parallel. I do not assert this; I test it through the leads in Equation (5) and probe it further in Section 4.5.

Why a common shock rather than staggered adoption: The Week 2 draft proposed exploiting cross-industry variation in AI *adoption timing*. A common single-date shock interacted with a continuous cross-sectional dose is the more defensible perimeter. With one treatment date shared by all units, the heterogeneity-and-timing pathologies of staggered two-way fixed-effects estimators (Callaway and Sant'Anna, 2021; de Chaisemartin and D'Haultfœuille, 2020; Goodman-Bacon, 2021; Sun and Abraham, 2021, among others) do not bind, because there are no forbidden comparisons between early- and late-treated cohorts. While this single-shock design avoids the pitfalls of staggered adoption, it carries its own strict identifying requirement: parallel trends across the entire exposure distribution. I formally test this assumption using the event-study leads and the robustness checks detailed below.

4.4 Data and Sample

The primary data source is the Survey of Income and Program Participation (SIPP), a true longitudinal panel that tracks occupation, education, and sub-annual job changes for the same workers over several years. The 2022–2025 SIPP waves (reference years 2021–2024) span the November 2022 shock; the estimation window is 2021m1–2024m12. The estimation universe is college-educated (bachelor's or above) workers aged 25–55 who are employed and correctly matched at baseline. SIPP is the only widely available US source that observes within-worker job-to-job transitions across the shock, which is what Equation (2) requires. The Current Population Survey[4] serves as a robustness sample for the cross-sectional dose-response: it is larger and timelier but follows workers only about sixteen months, which weakens identification of transitions across the shock. Three crosswalks support the construction: Eloundou scores to SOC, O*NET Job Zones to SOC, and SIPP/CPS occupation codes to SOC.[5]

4.5 Threats to Identification

I group the leading threats under three headings and state how the design answers each. Several are not fully eliminable, and where that is so I say what the residual bias does to β.

1. **Validity of the counterfactual:** Three forces could break the parallel-trends assumption. First, high-exposure occupations may already be on distinct trajectories, whether from the 2020–2021 remote-work reallocation or the pre-ChatGPT deterioration Frank et al. (2026) date to early 2022. The event-study leads in equation (5) test this directly, and I probe it further with occupation- or industry-specific linear trends. Second, the November 2022 date coincides with the 2022–23 technology-sector layoff and hiring-freeze wave, a confound contemporaneous with t^* that the leads cannot rule out. I therefore condition on industry layoff and vacancy intensity such as WARN notices and job-posting data, and I read β as not fully separable from this cycle where the controls leave residual variation. As falsification, I assign a placebo shock in November 2020 and re-estimate the model on low-exposure manual occupations, expecting $\beta \approx 0$ in both. Given that Frank et al. (2026) identify occupational deterioration in early 2022, I evaluate any non-zero placebo effects alongside the event-study leads. Pre-trends that scale systematically with exposure are netted out as early AI dynamics, whereas any pre-trend unrelated to exposure is treated as a genuine identification threat.

[4] U.S. Census Bureau, available at: `https://www.census.gov/programs-surveys/cps.html`; harmonized microdata via IPUMS-CPS at `https://cps.ipums.org`.

[5] The Standard Occupational Classification (SOC) system and its crosswalks are maintained by the U.S. Bureau of Labor Statistics. Available at: `https://www.bls.gov/soc/`.

2. **Sample selection and the control group:** The outcome is observed only for workers who stay employed and in the SIPP panel. So exposed workers may exit employment, or they may drop out of the panel non-randomly, and either would bias β. The thesis predicts downgrading rather than exit, so this margin should be quiet. I check it by running employment and labor-force participation as parallel outcomes (these should stay flat), and by comparing attrition across the exposure distribution. Where residual selection is non-negligible, I report Lee (2009) bounds on β. A second problem runs the other way. The cascade sends displaced high-exposure workers *into* low-exposure occupations, so the comparison group is itself partly treated and β is biased toward zero. I read the estimate as a conservative lower bound and, as a sharper alternative, compare the highest-exposure workers against the manual and in-person occupations least likely to receive downgraders.

3. **Measurement of treatment and outcome:** The Eloundou score captures potential exposure, not realized use, so β is an intention-to-treat parameter. I re-estimate with Anthropic Economic Index usage shares and treat agreement in sign and rough magnitude as the robustness criterion, and I guard against exposure merely proxying pre-existing restructuring by conditioning on routine-task and offshorability indices. On the outcome side, the realized-matches benchmark can manufacture overeducation by construction, so the externally built O*NET Job Zone classification serves as an independent check. Because occupation miscoding and crosswalk error inject non-classical misclassification into both the outcome and the dose, I report sensitivity to occupation-cell-size thresholds.

5 Contribution

I state three contributions, each tied directly to the design above.

1. **Empirical:** The project would provide one of the first within-worker causal estimates of AI-attributable credential–occupation mismatch among incumbent knowledge workers, identified off the ChatGPT shock interacted with occupational exposure. This closes the specific empirical gap—no link from occupational AI exposure to the credential match of subsequent employment—that currently allows the Massenkoff and McCrory (2026) stability finding to be read as resilience. The design is built so that a sufficiently precise null is as informative as a positive effect.

2. **Measurement:** The project would deliver a reproducible credential-mismatch index built from public survey and classification data. If the estimate is positive, the index reframes a flat unemployment signal as a measurement misalignment rather than an

all-clear; if it is null, the same index bounds how large any such misalignment could be.

3. **Practical:** The mismatch-transition rate by occupational exposure, computed from the timely and widely available CPS, would furnish a single leading indicator of within-employment labor erosion—a margin the current monitoring infrastructure does not track. Where the headline unemployment rate registers displacement only once it culminates in job loss, this indicator would surface the quality-margin adjustment in near real time, giving analysts and policymakers an early-warning measure that the standard dashboard omits.

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